Drones extend the wellsite control platform.

Fresh visual context helps operators see low-producing and hard-to-reach assets, then act on the conditions most likely to affect barrels, cost, and realized value.

+48 bbl/mo
ILLUSTRATIVE RECOVERED PRODUCTION

$2.7k/mo
TRUCK-ROLL COST AVOIDED

+$1.50/bbl
NETBACK TIMING ASSUMPTION

Synthetic 20-well example · all numbers are adjustable assumptions, not observed performance or a forecast.

PRODUCTION RESPONSE

Find the lost barrels sooner

The drone flags a likely issue; operator review and a fast work order shorten avoidable downtime.

20
STRIPPER WELLS

8 bopd
AVERAGE PER WELL

4/mo
EARLY FINDINGS

Each early finding avoids an assumed 1.5 days offline. At 8 bopd, four events recover 48 barrels per month.

4 × 1.5 × 8
EVENTS × DAYS × BOPD

48 bbl/mo
ABOUT 1% OF 4,800 BBL BASELINE

At an illustrative $70/bbl gross oil value, recovered production is $3,360/month before royalties, taxes, variable production costs, and any response expense.

Use inspection context to avoid low-value trips

Route people when the evidence warrants a visit; avoid trips when a remote check can resolve the question.

6
TRUCK ROLLS AVOIDED / MONTH

$450
ASSUMED COST / ROLL

$2,700
MONTHLY COST AVOIDED

The same scenario assumes two drone patrol sorties per month at $600 each. Patrol cost is $1,200/month, leaving $1,500/month in modeled truck-versus-patrol savings.

First principle

Compare the expected cost of action with the expected cost of waiting. Dispatch when expected loss, risk, or lost production exceeds the trip and repair cost.

Truck-roll savings do not include repair labor, parts, regulatory response, or avoided major failures; those benefits are not counted in this example.

Drones add context; connected data enables netback action

Aerial imagery is periodic or on demand. Continuous SCADA and commercial signals provide the real-time operating picture.

DRONE + SCADA → **WELL + INVENTORY** → **MARKET + LOGISTICS** → **OPERATOR RESPONSE**

visual anomalies · alarms → rate · tank · capacity → demand · price · nominations → produce · store · dispatch

Illustrative case: 4,800 baseline bbl/month × 40% volume with timing flexibility × $1.50/bbl improvement = $2,880/month modeled netback uplift.

This value requires current buyer / price signals, nominations, storage, and transport constraints in the decision loop. A drone alone does not change realized price.

Keep the action bounded by operating limits and human approval. Record recommendation, approval, dispatch, confirmed cause, and actual sales realization.

ILLUSTRATIVE ECONOMICS

The synthetic case models $7,740/month after patrol cost

20 wells · 8 bopd/well · 30-day month · scenario values are editable in the demo.

Recovered oil value	48 bbl × $70/bbl	**$3,360**
Avoided truck rolls	6 × $450	**$2,700**
Netback timing uplift	4,800 bbl × 40% × $1.50	**$2,880**
Gross modeled value	Sum of modeled benefits	**$8,940**
Drone patrol expense	2 sorties × $600	**–$1,200**
Net modeled opportunity	Before royalties, taxes, variable costs	**$7,740**

$92,880

ANNUALIZED NET OPPORTUNITY

Synthetic only. Excludes royalties, taxes, variable lift / repair costs, decline, downtime overlap, drone program overhead, and value of unmodeled risk. Validate each input against a measured pilot baseline.